

up funded by NASA, has developed a 3D printer that can print pizzas with edible ink. Astronauts need to choose the size, dough, topping and cheese, and the printer gets to work depositing layer after layer of 'ink' to create a pizza which is then baked in a 400-degree pizza oven. The pizza gets ready quite fast with minimal user intervention, so the company hopes that eateries on Earth will also utilise their tech!

The ultimate in JIT inventory

Moving beyond the hobbyists' zone, 3D printing is now well-rooted in industrial spaces too. The ability to manufacture the required parts just when these are needed could mean the next generation of just-in-time (JIT) inventory management. Automakers and others would no longer have to depend on vendors to meet their commitments.

Volkswagen Autoeuropa, an automobile manufacturing facility, has seen such benefits by using Ultimaker's desktop 3D printers to produce custom tools and parts, like gauges, jigs and fixtures. This is just a desktop 3D printer and not a large-scale industrial one, yet the benefits are clear. In a press report, Autoeuropa shared how they were able to 3D print a wheel-protection jig for around \$20. Had they gone to a vendor, the process could have cost more than \$850. By printing around a thousand parts, they claim to have saved \$160,000 in 2016.

Apart from reducing cost and improving customisation and productivity, 3D printing allows unlimited and on-demand design revisions, faster lead times and increased accessibility. Since only desktop 3D printers have been used by Volkswagen, the investment is quite low compared to the savings. It is also easy for workers to learn, and quite scalable.

When it comes to mass manufacture of plastic parts, experts feel that 3D printing is not as economi-

cal as injection moulding. So, it is likely to be used only for customised parts. However, when it comes to printing metal parts, it could be a game changer.

At CES 2017, Greg Mark, CEO of Markforged, showed off the Metal X desktop printer, which can print us-



Markforged's Metal X 3D printer can print metal parts overnight (Courtesy: Markforged)

ing metals like 17-4 stainless steel, 303 stainless steel, 6061 aluminium, 7075 aluminium, A-2 tool steel, D-2 tool steel, IN alloy (Inconel) 625 and titanium Ti-6Al-4V. The designs are printed in metal powder surrounded by plastic, then the plastic is dissolved and the metal sintered, leaving behind a strong metal part.

Metal X uses a technology called Atomic Diffusion Additive Manufacturing (ADAM), which, the company says, is at the intersection of 3D

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